



Experiment 11: Implement Normalization on the table in a database

Aim: Implement Normalization on the table in a database.

Theory:

Database normalization is a technique for designing relational database tables to minimize duplication of information and, in so doing, to safeguard the database against certain types of logical or structural problems, namely data anomalies.

For example, when multiple instances of a given piece of information occur in a table, the possibility exists that these instances will not be kept consistent when the data within the table is updated, leading to a loss of data integrity.

A table that is sufficiently normalized is less vulnerable to problems of this kind, because its structure reflects the basic assumptions for when multiple instances of the same information should be

Normalization is a process of converting a relation to be standard form by decomposition a larger relation into smaller efficient relation that depicts a good database design.

- 1NF: A Relation scheme is said to be in 1NF if the attribute values in the relation are atomic.i.e., Multi-valued attributes are not permitted.
- 2NF: A Relation scheme is said to be in 2NF,iff and every Non-key attribute is fully functionally dependent on primary Key.
- 3NF: A Relation scheme is said to be in 3NF,iff and does not have transitivity dependencies. A Relation is said to be 3NF if every determinant is a key for each & every functional dependency.
- BCNF: A Relation scheme is said to be BCNF if the following statements are true for eg FD $P \rightarrow Q$ in set F of FDs that holds for each FD. $P \rightarrow Q$ in set F of FD's that holds over R. Here P is the subset of attributes of R & Q is a single attribute of R represented by a single instance only.

Normalized tables are:- `mysql> create table Bus(BusNo varchar(20) primary key,Source varchar(20),Destination varchar(20));`

`mysql> Create table passenger(PPN varchar(15) Primary key,Name varchar(20),Age integer,Sex char,Address varchar(20));`

`mysql> Create table PassengerTicket(PPN varchar(15) Primary key,TicketNo integer);`

Conclusion: To be written by the students.

1. Explain the need of normalization?
2. What is functional dependency?
3. Explain difference between third normal form and boyce codd normal form?
4. What is PJNF?
5. What is transitivity dependency?